



UNIVERSITÀ
DI TORINO



MSSC2025
TORINO 8-12 SEP 2025

TODAY'S PROGRAM

MONDAY 08TH

AFTERNOON ADVANCED TUTORIALS

Copernico Room

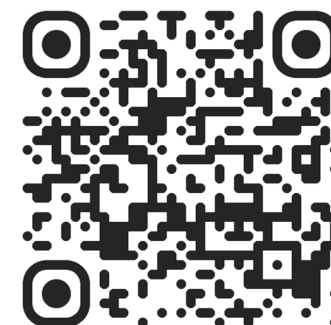
- 14:30 - 15:00 **Hands-On Kickoff** (Giacomo Ambrogio)
- 15:00 - 16:00 **How to Run CRYSTAL in Parallel** (Giacomo Ambrogio)
- 16:00 - 16:30 **Coffee Break**
- 16:30 - 17:15 **One-Electron Properties** (Laura Bonometti)
- 17:15 - 18:00 **QTAIM** (Marcos Rivera)

AFTERNOON BASIC TUTORIALS

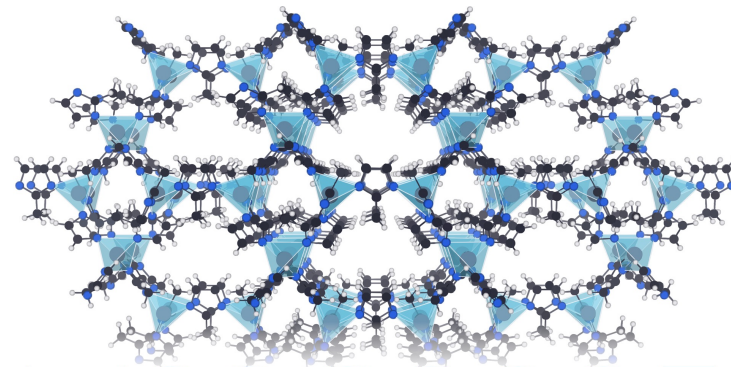
von Neumann Room

- 14:30 - 15:00 **Hands-On Kickoff** (Davide Mitoli)
- 15:00 - 16:00 **Geometry Input** (Andreha Gelli)
- 16:00 - 16:30 **Coffee Break**
- 16:30 - 18:00 **Basis Set, SCF and Single-Point Energy** (Andreha Gelli)

View the full
timetable online



Ab Initio Modelling in Solid State Chemistry



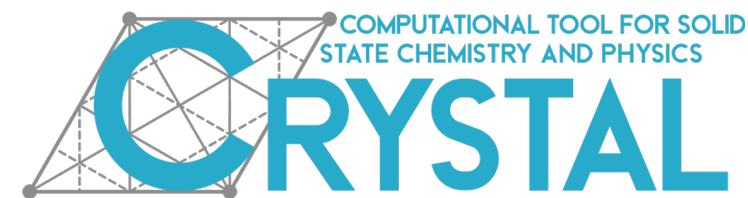
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Hands-On Kickoff

Davide Mitoli
(davide.mitoli@unito.it)

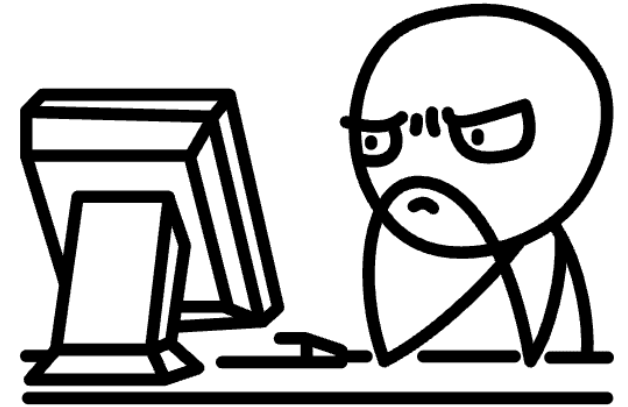


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Let's Get Started

- ✓ Afternoon tutorials focus on a **hands-on** approach.
- ✓ Topics:
 - ✓ Single-point (energy)
 - ✓ Geometry optimization and EoS
 - ✓ Modelling of 2D systems
 - ✓ Vibrational frequencies and spectra
 - ✓ One-electron properties
 - ✓ Basics of TOPOND (electron density topology analysis)
 - ✓ Response properties (CPHF/KS)
 - ✓ Modelling of defects



Virtual Machine

- ✓ You should all have installed the MSSC2025 virtual machine (VM). If not, please install it now. If you run into any problems, ask the tutors in the room or in the chat.
- ✓ The VM ships with a pre-configured environment that includes everything you need for the afternoon tutorials:
 - ✓ pre-compiled binaries (CRYSTAL and PROPERTIES)
 - ✓ running scripts
 - ✓ links to useful webpages
 - ✓ input and output files (MSSC2025 folder, GitHub repository)
- ✓ Debian GNU/Linux 12
 - ✓ password: `crystalcode`



The Command Line in a Nutshell

<code>pwd</code>	Display path of present working directory
<code>cd <dir></code>	Change directory to <code><dir></code>
<code>ls</code>	List directory content
<code>mkdir <new_dir></code>	Create new directory named <code><new_dir></code>
<code>mv <file> <dir></code>	Move <code><file></code> to <code><dir></code>
<code>cp <file> <dir></code>	Copy <code><file></code> to <code><dir></code>



You can find a cheat sheet in `MSSC2025/Basic/Day1/KickOff`

Running

CRYSTAL

- ✓ Core features
- ✓ Input file: `input.d12`
- ✓ How to run:
 - ✓ `runCRY23 <input>`
or (for restarting calculations)
 - ✓ `runCRY23 <input> <restart_file>`

PROPERTIES

- ✓ Post-processing
- ✓ Input file: `input.d3`
- ✓ How to run:
 - ✓ `runPROP23 <input> <CRYSTAL_output>`

File extensions (like .d12 or .d3) must be omitted!

You can find a cheat sheet in `MSSC2025/Basic/Day1/KickOff`

MSSC2025 GitHub Repository

- ✓ All files for the afternoon tutorials are available on GitHub at <https://github.com/crystaldevs/MSSC2025>
- ✓ A local copy of the repository is already available on your VM.

- ✓ To fetch the latest changes:

```
cd MSSC2025
```

```
git pull
```

- ✓ If anything goes wrong, just delete the folder and clone the repository again — it's a quick fix!

```
rm -r MSSC2025
```

```
git clone https://github.com/crystaldevs/MSSC2025
```



Visualization Tools

- ✓ CRYSPLIT is a user-friendly web-based tool for quick and easy visualization
 - ✓ Upload output files and visualize charts directly on your browser
 - ✓ Libraries: plotly, jsmol



- ✓ CRYSTAClear is a python package that handles I/O with CRYSTAL and offers advanced plotting methods:
 - ✓ Libraries: numpy, matplotlib, pandas, pymatgen, ...
 - ✓ Jupyter notebook on Google Colab
 - ✓ In-development

